



Fig. 3: A child solving a puzzle using a touchscreen (Courtesy: Wikipedia)



Fig. 4: Snapshots from the video of answering an incoming call with the help of gestures

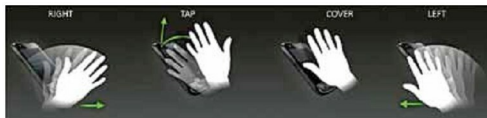


Fig. 5: Various gestures demonstrating simple applications in a mobile phone (Courtesy: www.eyesight-tech.com)



Fig. 6: Zooming in and out using gesture recognition technology (Courtesy: www.pointgrab.com)

gloves. Mobile operating systems like Android also find their way to a big screen like TV's. With that, this new experience will be offered to a broader audience on a wider range of devices. Controlling a TV using hands without the need for a remote control is the kind of freedom consumers are looking for.

"Interacting with consumer electron-

ics is about to take another giant leap forward as hand gesture recognition is increasingly being incorporated into our smartphones, tablets, laptops, all-in-ones and more," says Assaf Gad, vice president of marketing for PointGrab. "It is an exciting time for both OEMs and end-users, as our devices are becoming even more intuitive and entertaining."

Korean phone maker Pantech, best known in the US for low-end Android and messaging phones, has added gesture-based controls to its Vega LTE handsets, providing Kinect-like gesture recognition in Android phones. By incorporating the recognition software, it allows users to control the handset with gestures. The technology, which comes from Israel's eyeSight Mobile Technologies, is a part of the Vega LTE line of phones.

According to eyeSight, gestures are useful when touch input is impractical, such as when driving or wearing

gloves. Using the front-facing camera and eyeSight's software, it helps smartphone owners to control their phones by waving their hand. Among the functions possible via gesture recognition will be answering calls and playing music.

"Natural hand gesture recognition is an up and coming user interface capability. With natural language recognition, such as Siri or Google Now, we are seeing a major user interface evolution similar to the one that occurred when touchscreens were introduced in mobile devices," says Amnon Shenfeld, eyeSight's vice president-R&D.

A YouTube video shows this technology in action, where a woman's hands are covered in batter and with the help of gesture recognition technology, she answers the incoming call. Snapshots from the video can be seen in Fig. 4.

To sum up

Touch is still a key user interface but with all the sensors in our smartphones, these devices can be expected to adopt what can be called the 'invisible interface.' Gesture recognition might seem a mismatch with phones since people tend to keep them close at hand, but it is believed that there are plenty of reasons why people will want to manage their handsets with movements.

For example, when a phone is positioned on a car dashboard, a driver could utilise gestures to answer or ignore an incoming call, activate voice recognition features, zoom in or out on a map (see Fig. 6) and adjust volume of the device. The exact gestures would vary based on the phone model or personal preferences but would likely include a series of waves, swoops, pats and flicks.

But it's not so simple...

There are a lot of people who question the need for touchless interaction on a device that is designed to be in our hand at all times! The use-cases for touchless gestures on a mobile device are incredibly few and far between. The only reason why touch-free control makes sense on smartphones or tablets is that "touch-free means smudge-free."

If your hands are dirty or you hate smudges, touch-free controls are a benefit. But beyond that there is little need for touch-less gesture controls. Hands are designed to touch and manipulate things. Moreover, a key limitation is that motions, such as a hand flick or circular movement, are recognised only within a 15cm (6-inch) range. Also, power consumption is a key issue for battery-powered devices.

Several challenges remain for gesture recognition technology on mobile devices, including effectiveness of the technology in adverse light conditions, variations in the background and high power consumption. However, it is believed that these problems can be overcome with different tracking solutions and new technologies. ●

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